# Sentinel-1 / Sentinel-2 land-cover classification of Porto

SNAP collocation, Random Forest, and why 100 percent accuracy is not a result

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github.com/aleksandrapacia/PortoProject

## 1. What this project is

A SNAP + Python workflow over Porto and the Douro valley. Sentinel-1 radar and Sentinel-2 optical images are collocated in SNAP, then a Random Forest in Python assigns each pixel to water, vegetation, urban, or bare soil. The point is not only the map. It is to fuse two sensors and to check whether a high accuracy score is actually meaningful.

## 2. Method

- SNAP: collocation of Sentinel-1 GRD (sigma0 VV, VH) with Sentinel-2; MNDWI and NDVI as auxiliary layers.
- Labels (v1): pixel thresholds on MNDWI, NDVI and VH. These are not independent ground truth.
- Classifier: Random Forest on raw bands only (B2, B3, B8, B11, sigma0 VV, sigma0 VH). Indices were removed from the feature vector after a 100 percent score showed leakage.
- Validation: stratified random split and spatial block cross-validation (GroupKFold on a 10x10 grid).

## 3. Results

A random split produced essentially perfect scores (water and vegetation F1 = 1.00; urban about 0.97; overall accuracy about 1.00). Spatial-block CV stayed near 0.99. That is still too high to believe as map accuracy: the labels remain threshold rules on the same image.

What is usable: the classified map is spatially coherent (ocean and Douro as water; built-up as fragmented red; vegetation dominant inland). Feature importance shows optical structure as the main driver, with VH backscatter contributing to the urban class. Radar is not redundant.

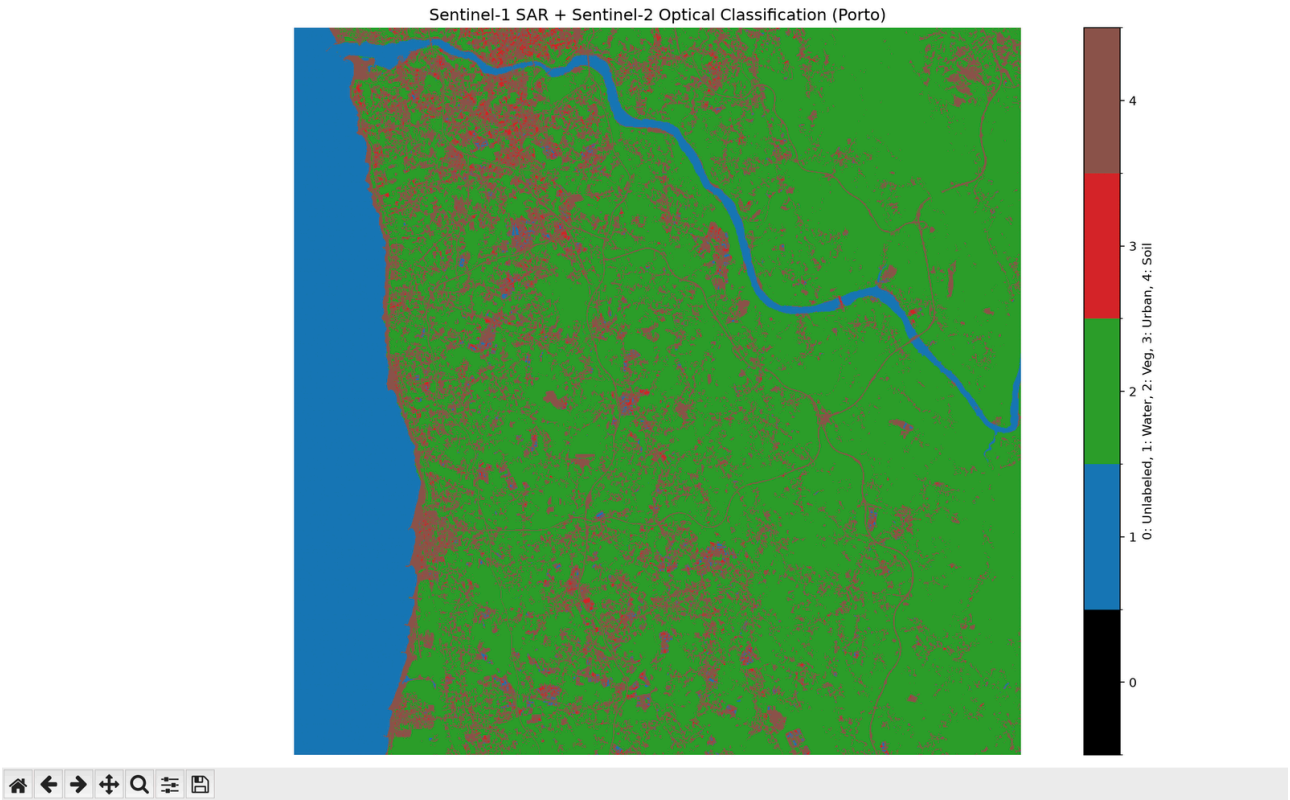



Figure 1. Random Forest map from SNAP-collocated S1+S2. Blue: water; green: vegetation; red: urban; brown: bare soil.

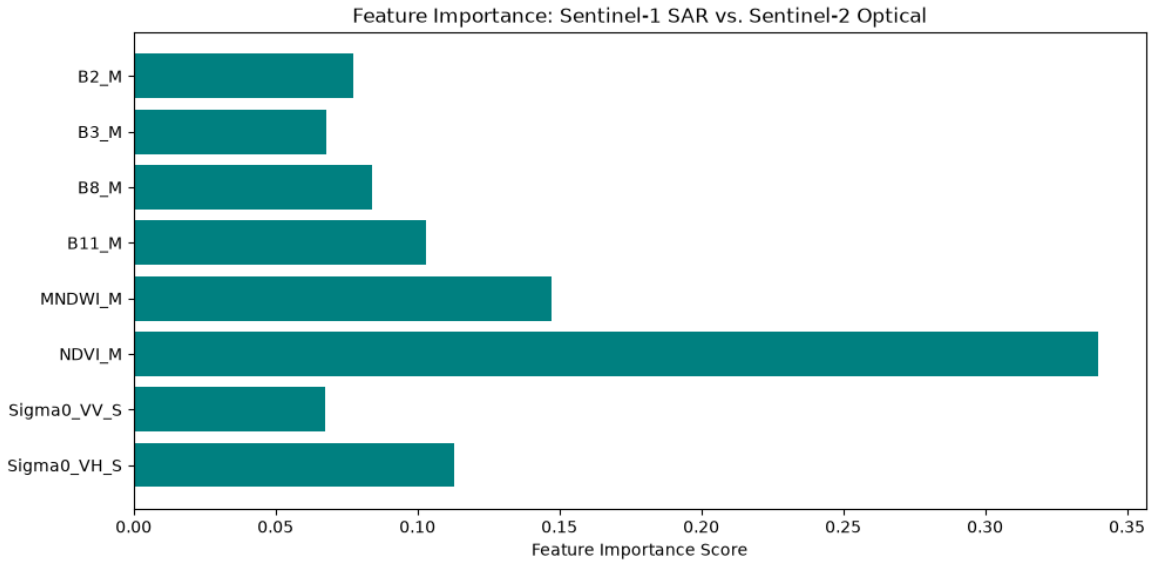



Figure 2. Feature importance (diagnostic run with indices). NDVI dominates; sigma0 VH is the strongest radar contributor.

## 4. Limitation

The 100 percent figure was a data leak: labels derived from NDVI/MNDWI, then a tree model reproducing that formula, plus neighbouring pixels in both train and test (spatial autocorrelation). Removing indices from training and using block CV is necessary but not sufficient. Independent polygons (hand ROIs, or CORINE / ESA WorldCover) are required before any accuracy is reported as a scientific result.

## 5. Possible next steps

- Independent test labels: SNAP/QGIS ROIs versus CORINE or ESA WorldCover.
- Add DEM derivatives (slope, aspect) for Douro valley relief.
- Same spatial CV, second classifier (SVM or XGBoost) for a method comparison.